Exercise 1

a) We claim that $\lim_{n\to+\infty} \frac{(-1)^n}{n} = 0$

We have to prove that $\forall \varepsilon > 0 \ \exists n_0 > 0$ s.t. $\forall n > n_0 \ \left|-\frac{(1)^n}{n} - 0\right| < \varepsilon$.

Notice that $\left|\frac{-(1)^n}{n}\right| = \frac{1}{n}$

By the Archimedean principle, given $\varepsilon > 0 \ \exists n_0$ s.t. $\frac{1}{n_0} < \varepsilon$

~~Then~~ $\forall n > n_0 \ \left|-\frac{(1)^n}{n}\right| = \frac{1}{n} < \frac{1}{n_0} < \varepsilon$ as whished.

b) $\lim_{n\to\infty} \frac{n}{n+1} = 1$.

We ~~know~~ have to examine $\left|\frac{n}{n+1} - 1\right| = \left|-\frac{1}{n+1}\right| = \frac{1}{n+1}$

As before (by Archimedean principle), given $\varepsilon > 0$ we pick $n_0 \in \mathbb{N}$ s.t. $\frac{1}{n_0} < \varepsilon$. Then $\forall n > n_0 \ \left|\frac{n}{n+1} - 1\right| = \frac{1}{n+1} < \frac{1}{n_0} < \varepsilon$ as desired.

c) $n^{(-1)^n}$ doesn't converge

Suppose $\lim_{n\to\infty} n^{(-1)^n} = x$, ~~then~~ Fix $\varepsilon = 1$

Then $\exists n_0$ s.t. $\forall n > n_0 \quad |n^{(-1)^n} - x| < 1$

But if we take n even, $n > n_0$, $n > |x| + 1$ (which is possible by archimedean principle) we find

$|n^{(-1)^n} - x| = |n - x| \geq |n| - |x| \geq 1$ contraddiction

d) $\lim_{n\to\infty} \frac{\sin(n)}{\sqrt{n}} = 0$

Notice that $0 \leq \left|\frac{\sin(n)}{\sqrt{n}}\right| \leq \frac{1}{\sqrt{n}}$

Fix $\varepsilon > 0$. We would like to determine for what $n \in \mathbb{N}$ we have $\frac{1}{\sqrt{n}} < \varepsilon$. It is equivalent to $\frac{1}{n} < \varepsilon^2$, which has a solution n_0, by the Archimedean principle. Then

$\left|\frac{\sin(n)}{\sqrt{n}}\right| \leq \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{n_0}} < \varepsilon \ \forall n > n_0$, as desired.

Exercise 2

a) Let $C > 0$ be a bound for $|a_n|$, which exist by definition.

Fix $\varepsilon > 0$. And let $\varepsilon' = \varepsilon / C$. By definition of limit, $\exists n_0$ s.t. $|b_n| < \varepsilon' \ \forall n > n_0$. Then, $\forall n > n_0$

$|a_n b_n| \leq |a_n||b_n| \leq C \cdot \varepsilon' = C \cdot \varepsilon / C = \varepsilon$ as desired.

b) Let $x = \lim_{n\to+\infty} x_n$. Fix $\varepsilon > 0$, and let $\varepsilon' = \varepsilon/2$.

By definition of limit $\exists n_0$ s.t. $|x_n - x| < \varepsilon' \ \forall n > n_0$.

Then $|x_{n+1} - x| < \varepsilon'$ holds as well if $n > n_0$.

For $n > n_0$ we then have $|x_{n+1} - x_n| \leq |x_{n+1} - x| + |x_n - x| \leq 2\varepsilon' = \varepsilon$.

$(x_{n+1} - x_n = (x_{n+1} - x) - (x_n - x))$ by triangular inequality.

Exercise 3

a) We treat three different cases, according to the sign of the arguments of the absolute values.

For $x \leq -1$ we get: $-x-\cancel{1}-x+\cancel{1} < 3$

$x > -\frac{3}{2} \wedge x \leq -1$

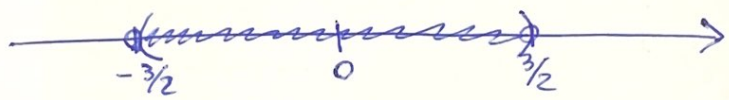
For $-1 < x < 1$: $\cancel{x}+1-\cancel{x}+1 < 3$

always true when $-1 < x < 1$

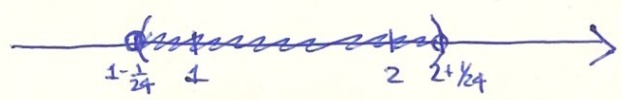
For $x \geq 1$: $x+\cancel{1}+x-\cancel{1} < 3$

$x < 3/2 \wedge x \geq 1$

Taking the union of the three sets of solution, we get that the set of solution is $\{x \in \mathbb{R} : -3/2 < x < 3/2\}$

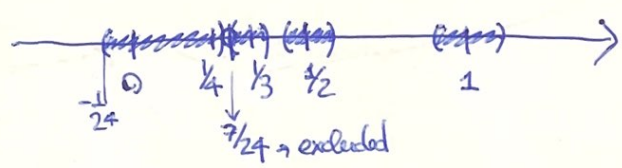


b) i)



$X_\varepsilon = (\frac{23}{24}, \frac{49}{24})$ notice that endpoints are excluded.

ii)



$X_\varepsilon = (-\frac{1}{24}, 7/24) \cup (7/24, 9/24) \cup (\frac{11}{24}, \frac{13}{24}) \cup (\frac{23}{24}, \frac{25}{24})$

c) By assumption $\exists\, x \in A_\varepsilon \cap B_\varepsilon$

We claim that $(x-\varepsilon, x+\varepsilon)$ is the desired interval

By definition of A_ε, since $x \in A_\varepsilon$ $\exists\, y \in A$ s.t. $|x-y| \leq \varepsilon$.

Then $y \in (x-\varepsilon, x+\varepsilon) \Rightarrow A \cap (x-\varepsilon, x+\varepsilon) \neq \emptyset$ (since it contains y).

By the same reasoning $(x-\varepsilon, x+\varepsilon) \cap B \neq \emptyset$ and we are done.

Exercise 4

2) • First we prove that $\alpha \leq \beta$. The other two inequalities follow from def. of sup and inf.
Suppose by contraddiction that $\beta < \alpha$.
Set $\varepsilon = |\beta - \alpha|$ and by ~~defin~~ the characterization of sup and inf we can find $\bar{a} \in \{a \in \mathbb{R} \mid [a,b] \in I\}$ s.t. $\alpha - \bar{a} < \varepsilon/2$, and ~~since~~ $\tilde{b} \in \{b \in \mathbb{R} \mid [a,b] \in I\}$ s.t. $\tilde{b} - \beta < \varepsilon/2$.
Hence by triangular inequality $\bar{a} > \tilde{b}$.
In particular we have $[\bar{a}, \bar{b}] \in I$, and $[\tilde{a}, \tilde{b}] \in I$, but these two intervals are disjoint since $\tilde{b} < \bar{a}$, and then not nested, giving us the desired contraddiction.

• Then $[\alpha, \beta]$ is non empty.

• We finally prove that $\bigcap_I [a,b] = [\alpha, \beta]$
For any $[a,b] \in I$ we have $a \leq \alpha$ and $\beta \leq b$ hence $[\alpha, \beta] \subset [a,b] \Rightarrow [\alpha, \beta] \subset \bigcap_I [a,b]$.
Let us prove the opposite containment.
We need "$x \in \bigcap_I [a,b] \Rightarrow x \in [\alpha, \beta]$". We'll prove the contrapositive "$x \notin [\alpha, \beta] \Rightarrow x \notin \bigcap_I [a,b]$". By De Morgan law it is equivalent to "$x \notin [\alpha, \beta] \Rightarrow x \in \bigcup_I [a,b]^c$".
Without loss of generality $x < \alpha$. Then set $\alpha - x = \varepsilon$
By def of sup we can find $\bar{a} \in \{a \in \mathbb{R} \mid [a,b] \in I\}$ s.t. $a - \bar{a} < \varepsilon$. Then $\bar{a} > x$. Therefore there exists $[\bar{a}, \bar{b}] \in I$ s.t. $x \notin [\bar{a}, \bar{b}]$. Hence $x \in \bigcup_I [a,b]^c$ as wanted.

• Since $\bigcap_I [a,b]$ contains $[\alpha, \beta]$, which is non-empty, it is nonempty itself.

6) We take $I = \{(0, \frac{1}{n}) : n \in \mathbb{N}, n \neq 0\}$.
Clearly if $x \leq 0$ $x \notin (0,1) \Rightarrow x \notin \bigcap_I (a,b) = \bigcap_n (0, \frac{1}{n})$
If $x > 0$ $\exists\, \bar{n}$ s.t. $\frac{1}{\bar{n}} < x$ by Archimedean principle.
Then $x \notin (0, \frac{1}{\bar{n}}) \Rightarrow x \notin \bigcap_I (a,b) = \bigcap_n (0, \frac{1}{n})$.

Then $\bigcap_I (a,b) = \emptyset$.

Exercise 5

a) Let us pick $r = 1$ and $s = x$.
We claim that $\mathbb{R}(x)$ is not Archimedean, and these r and s are a counterexample to the fact that it is Archimedean.
We have to prove that $\forall\, n \in \mathbb{N}$ $\quad n \cdot r < s$, i.e. $n < x$,
which is true if and only if $\frac{x-n}{1} > 0$, as a rational function
Here $m=1$, $n=0$, $a_m = 1$, $b_n = 1$ $\rightarrow$ $a_m/b_n > 0$ and the claim is true.

b) If $\mathbb{R}(x)$ was complete then it would be isomorphic to $\mathbb{R}$ (being an ordered field). In this case it would be Archimedean. Contradicting part a).

c) As in a) we prove that x, as a rational function, is an upper bound for $\mathbb{R}$, so the set of upper bounds for $\mathbb{R}$ is not empty. Let us call it Q.
Suppose Q has a minimum q.
Since $0 \in \mathbb{R}$, $q > 0$.
We notice that if $q > c$ $\forall\, c \in \mathbb{R}$ then $q/2 > c/2$ $\forall c \in \mathbb{R}$. $c/2$ ranges in all $\mathbb{R}$, hence $q/2 \in Q$ as well
But since $q > 0$, $q/2 < q$ contradicting the minimality of q in Q.